Worksheet 5

Time series – EXERCISES

Badr ABOUFARASSE, PhD. – Academic year 2026–2027

Exercise 1 – Components and time-series models

Identify the component T, C, S, or R for: a general trend, a recession-related movement, a recurring quarterly peak, and an exceptional unpredictable shock. Write the complete additive and multiplicative models. State the simplification used later in the course when cyclical and residual components are ignored.

Exercise 2 – Centred moving average of order 4

For quarterly sales $18, 26, 20, 35, 25, 36, 28, 48$, calculate all simple 4-term moving averages, then centre them to obtain centred moving averages at periods 3 to 6. Explain why an even-order moving average must be centred.

Exercise 3 – Multiplicative decomposition and quarterly forecast

Quarterly sales (hundreds of euros) are:

Year	Q1	Q2	Q3	Q4
2012	18	26	20	35
2013	25	36	28	48
2014	38	52	40	65

Use the multiplicative model and trend $T(t) = 3t + 15.40$, with $t = 1$ for Q1 2012. Calculate the first two seasonal ratios, explain quarterly seasonal averaging, correct the rounded seasonal means $0.90, 1.15, 0.81, 1.25$ so that their mean is 1, and forecast 2015 ($t = 13$ to 16), both deseasonalised and seasonalised.